From Show and Tell to Attention and Transformers: An Empirical Study of Architecture for Image Captioning

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Abstract

Image captioning requires effective integration of visual understanding and natural language generation. In this work, we propose a unified image captioning framework for controlled comparison of visual representations and decoding architectures. A pretrained ResNet-50 [3] CNN encoder extracts global and spatial image features, which are combined with LSTM, attention-based LSTM, and Transformer decoders to form four model variants. The classical Show and Tell [11] model is used as the baseline, while the other variants introduce spatial attention and Transformer-based decoding [9]. Experiments on the Flickr8K [4] dataset using standard metrics (BLEU, METEOR, ROUGE-L, CIDEr) show that spatial features improve visual grounding, and Transformer decoders further enhance performance by better modeling contextual dependencies.

1. Introduction / Background / Motivation

1.1. Project Objective: Automatic Image Description

The goal of this project is to build a system that can look at a photograph and automatically generate a natural, human-like sentence describing what is happening in it. We aim to compare several different system designs under the exact same setup to identify which one writes the most accurate and descriptive captions.

1.2. Current Practice and Key Limitations

Current image captioning models usually use an image encoder and a text decoder, using attention mechanisms [12] to focus on important image regions. However, these systems have limitations; they frequently produce generic or inaccurate captions, miss fine visual details, and are sensitive to internal architecture choices such as hidden size, layers, attention design, dropout, learning rate, and decoding strategy.

1.3. Impact and Motivation

Solving this problem matters because countless images online lack useful text descriptions. A successful automated model greatly improves internet accessibility for visually impaired users, enhances image search capabilities, and helps organize large visual datasets. Furthermore, it makes image data much more useful in downstream applications that rely on combining both visual and textual information.

1.4. Dataset Details

All experiments were conducted using Flickr8k dataset [4], which is partitioned into 6k training images, 1k validation images and 1k test images. While larger datasets exist for image captioning task, Flickr8k was specifically selected to achieve an optimal balance between achieving meaningful model performance and maintaining manageable computation times during the iterative hyperparameter tuning phase.

2. Approach

2.1. Overall Framework

Figure 1 illustrates the unified image captioning framework, which follows a standard encoder-decoder paradigm. A pretrained ResNet-50 [3] is used as the shared visual encoder to extract image representations, while a sequence decoder generates captions in an autoregressive, word-by-word manner. From the encoder, two types of visual features are derived: a global feature obtained via global average pooling, and spatial features extracted from the final convolutional feature map that preserve region-level in-

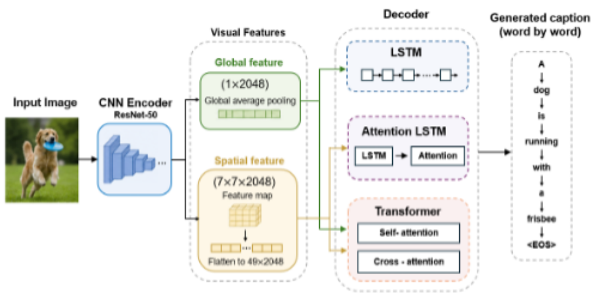


Figure 1. Overview of the unified image captioning framework with interchangeable encoder and decoder components.

Model	Encoder	Decoder	Description
M1	Global	LSTM	Show and Tell baseline
M2	Spatial	Attn-LSTM	Attention-based LSTM
M3	Spatial	Transformer	Transformer w/ spatial
M4	Global	Transformer	Transformer w/ global

Table 1. Model Configurations.

formation. On the decoder side, we consider three architectures: LSTM, attention-based LSTM, and Transformer. By combining different visual representations with different decoders, we construct four model variants (M1 to M4), as summarized in Table 1. The global feature + LSTM model (M1) corresponds to the Show and Tell [11] framework and serves as our baseline, while the other variants introduce spatial attention or Transformer-based decoding.

This unified design enables controlled comparisons along two key dimensions: visual representation (global vs. spatial features) and decoding architecture (LSTM, attention-based LSTM, and Transformer), allowing us to systematically evaluate their impact on caption generation performance.

2.2. Visual encoding

We use a pretrained ResNet-50 [3] as the shared visual encoder for all models. Show-and-Tell [11] originally used an Inception-based encoder; we adopt ResNet-50 for its stronger representations, broad adoption, and modular compatibility across model variants. ImageNet [2] pretrained weights are used with the backbone frozen during training.

Two visual representations are derived from the same backbone. **Spatial encoding** uses the output of the final convolutional block before global average pooling, producing a $7 \times 7 \times 2048$ feature map that is reshaped into 49 region features and projected to $(49, 512)$ for attention-based decoding. **Global encoding** is derived from the same spatial representation by average-pooling the 49 projected region features into a single 512-D image vector. The two representations capture complementary information: global features encode holistic scene semantics, while spatial features preserve region-level information for object-word alignment. Spatial features are expected to benefit caption quality through finer visual grounding, whereas global features provide a simpler, strong baseline.

2.3. Caption decoding architectures

All models formulate caption generation as autoregressive next-token prediction conditioned on image features and previously generated words. Three decoder families are compared:

- **LSTM (M1)** follows the encoder-decoder formulation of Show and Tell [11]: a global image feature initializes the recurrent decoder's hidden state, and tokens are generated step by step.
- **Attention-LSTM (M2)** extends the recurrent decoder with soft visual attention [12] over the 49 spatial regions at each step, enabling dynamic region selection during generation.
- **Transformer (M3, M4)** replaces recurrence with stacked masked self-attention and cross-attention layers [9]. Caption tokens attend both to previous words (causal self-attention) and to encoded visual regions (cross-attention). M3 uses spatial visual features; M4 uses the same global feature as M1, isolating the effect of decoder family under a fixed visual representation.

These architectures span increasing modeling capacity, from sequential recurrence, to recurrence with attention, to fully attention-based decoding, enabling a controlled comparison of decoder design under shared training and evaluation settings.

2.4. Word Embeddings

Three word embedding strategies are considered: **random initialization**, **pretrained GloVe [8] embeddings with frozen weights**, and **pretrained GloVe embeddings with fine-tuning**. In the first setting, embeddings are learned from scratch during training, serving as a baseline without external linguistic knowledge. In the second setting, embeddings are initialized using pretrained GloVe vectors (trained on large-scale text corpora to capture global word co-occurrence statistics) and kept fixed, providing stable semantic representations while reducing overfitting. In the third setting, pretrained embeddings are further fine-tuned, allowing adaptation to the captioning task and potentially improving alignment between visual and textual features.

The pretrained GloVe embeddings are 300-dimensional, while the model embedding size is 512. To address this mismatch, a linear projection layer is applied to map the embeddings into the model space. To ensure fair comparison, all models share the same vocabulary and embedding dimensionality, and only the embedding initialization and trainability are varied. Pretrained embeddings are expected to provide improvements over random initialization, while the additional benefit of fine-tuning may be limited due to the limited size of the training data.

2.5. Evaluation Metrics

Caption quality is evaluated using four standard metrics: **BLEU** [7], **METEOR** [1], **ROUGE**-L [5], and **CIDEr** [10], which are widely adopted in prior image captioning studies. BLEU measures n-gram precision between generated captions and reference captions, emphasizing local word overlap. METEOR incorporates both precision and recall with additional alignment mechanisms such as stemming and synonym matching, making it more sensitive to semantic similarity. ROUGE-L is based on the longest common subsequence and captures sentence-level structure and fluency. CIDEr measures consensus between generated captions and multiple references using TF-IDF weighting, emphasizing the importance of informative and distinctive words.

Among these metrics, CIDEr is considered the primary metric in this work, as it is specifically designed for image captioning and better reflects human judgment by accounting for consensus across multiple references. BLEU and ROUGE-L provide complementary insights into lexical overlap and structural similarity, while METEOR offers additional sensitivity to semantic variations. All metrics are reported to provide a comprehensive evaluation of model performance.

[TODO] Per the rubric, the report must mention any code repositories or other sources used and what changes were made. I referenced some of my code from Assignment 2 and 3, but did not reuse anything verbatim. Our GitHub Repo

3. Experiments and Results

3.1. Experimental setup and model configurations

Dataset. Flickr8k [4] (partition into training-6k, dev-1k, and test-1k data).

Loss function. All models were trained to minimize the cross-entropy loss, which calculates the per-token probability error of the generated caption against the ground truth.

Optimizer. The AdamW [6] optimizer was preferred over standard Adam. AdamW decouples weight decay from the gradient update step, which is recommended for training complex sequence models like transformer and LSTM, leading to better generalization.

Model	LR	WD	Hid.	Lyrs.	Heads	Drop.
M1	1e-4	0.05	512	2	NA	0.1
M2	1e-4	0.05	512	1	1	0.1
M3	1e-4	0.05	512	4	8	0.3
M4	1e-4	0.05	512	4	8	0.3

Table 2. Model-specific best hyperparameters. WD: weight decay; Hid.: hidden/dim; Lyrs.: layers; Drop.: dropout. All models are trained for 10 epochs with batch size 16.

Transfer Learning. Due to smaller dataset (Flickr8k), transfer learning was utilized to accelerate convergence across all the models. The weights of the Resnet-50 encoder were strictly frozen during training. The model primarily focuses on optimizing the decoder's ability to map established visual representations to text.

Hyperparameter Tuning Strategy. The hyperparameters were refined through iterative tuning, with specific attention to regularization mainly due to relatively small size of Flickr8k dataset use (Table 2):

- **Batch size:** A batch size of 16 was utilized. During tuning, smaller batch size provided a regularization effect through slightly noisier gradient updates, which helped the models escape sharp local minima.
- **Learning Rate Scheduler:** A learning rate scheduler was implemented, utilizing a warmup phase (first 2 epochs) followed by annealing. The warmup phase was to prevent early divergence, especially for transformer decoder, while the annealing allowed the optimizer to smoothly settle into the final optimal weights in the last few epochs.
- **Regularization:** to combat overfitting on the small dataset, a dropout rate ranging from 0.1 to 0.3 were applied to the encoder and decoder models respectively. And a weight decay of 0.05 was enforced via the AdamW optimizer to penalize overly large network weights.

Training results. Figure 2 illustrates the loss curves of two examples (M1 and M3) across 10 epochs. The baseline model (global + LSTM) demonstrates stable convergence, with training and validation loss aligning closely after 10 epochs. Conversely, despite aggressive regularization, the more complex Transformer exhibited a divergence between training and validation loss, which is typical for high-capacity models on small datasets. However, the Transformer ultimately achieved better overall performance compared to LSTM.

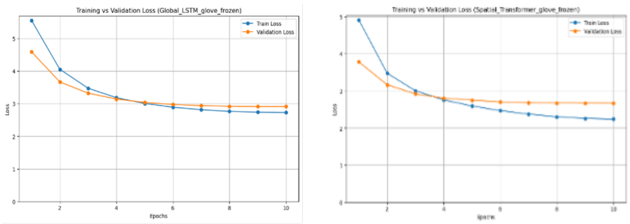


Figure 2. Training convergence comparison between M1 (left) and M3 (right).

3.2. Architecture Comparison

This section evaluates the four model variants to analyze how specific architectural choices influence caption quality. To ensure optimal sequence generation, all evaluation metrics were calculated using beam search decoding strategy. Table 3 summarizes the quantitative results for each model.

A consistent observation across the result is the impact of Visual Representation: Global vs Spatial (M1 vs. M2 & M4 vs. M3): preserving spatial features significantly improves performance for both LSTM and transformer based decoders.

- (M1 vs. M2): the baseline model forces the entire image into a single global vector, discarding localized visual details. By transitioning to CNN spatial features paired with an Attention-LSTM (M2), the decoder has the ability to focus on specific image regions for each generated word. This localized visual grounding generates higher caption accuracy of CIDEr score from 0.37 to 0.42.
- (M3 vs. M4): the transformer decoder shows a similar reliance on spatial information. The global feature discards the visual region details that cross-attention mechanisms in transformer decoder are designed to exploit. It results in more generalized and less accurate captions. On the other hand, with spatial feature map provided, the transformer can align distinct image regions with visual tokens, allowing the generated text to be tightly grounded to specific objects and backgrounds. As a result, M3 pushes the CIDEr score to a project-high of 0.47.

For the impact of decoder architecture, M3 outperforms M2 because the Transformer decoder uses the same spatial image features more effectively than the Attention-LSTM decoder. In M2, caption generation depends on a recurrent hidden state, so visual and language context must be carried forward step by step. This can limit how well the decoder preserves earlier information, especially when generating longer phrases. In contrast, M3 uses masked self-attention, allowing each word position to directly reference previous caption tokens, while cross-attention links those tokens to spatial image regions. This structure gives M3 a more flexible way to model object relationships, phrase structure, and image-text alignment. Therefore, with the same spatial encoder input, M3 produces more coherent and better-grounded captions than M2.

Model	B-1	B-2	B-3	B-4	MET.	R-L	CIDEr
M1	0.56	0.37	0.24	0.15	0.36	0.44	0.37
M2	0.58	0.40	0.26	0.17	0.37	0.46	0.42
M3	0.58	0.40	0.27	0.18	0.38	0.46	0.47
M4	0.57	0.39	0.25	0.17	0.37	0.45	0.43

Table 3. Quantitative architecture comparison (best hyperparameter model). B-n: BLEU-n, MET.: METEOR, R-L: ROUGE-L.

Embedding	B-1	B-2	B-3	B-4	MET.	R-L	CIDEr
Random	.568	.379	.245	.155	.355	.445	.370
GloVe frozen	.562	.372	.241	.152	.356	.442	.373
GloVe finetune	.558	.367	.237	.155	.355	.445	.370

Table 4. Metrics results for baseline M1 (Global+LSTM).

Embedding	B-1	B-2	B-3	B-4	MET.	R-L	CIDEr
Random	.546	.369	.243	.154	.368	.454	.417
GloVe frozen	.572	.386	.255	.165	.368	.452	.421
GloVe finetune	.573	.389	.260	.169	.370	.455	.422

Table 5. Metrics results for M2 (Spatial+LSTM Attention).

Embedding	B-1	B-2	B-3	B-4	MET.	R-L	CIDEr
Random	.581	.398	.271	.181	.383	.465	.473
GloVe frozen	.582	.398	.271	.183	.379	.463	.467
GloVe finetune	.587	.404	.275	.184	.380	.463	.478

Table 6. Metrics results for M3 (Spatial+Transformer).

3.3. Embedding Comparison

To assess the impact of prior semantic knowledge on the image caption generation process, three models, baseline (M1: Global+LSTM), M2 (Spatial+Attention-LSTM) and M3 (Spatial+Transformer), were evaluated across three word embedding configurations. Although the experimental results in Tables 4, 5, and 6 demonstrate that the introduction of pre-trained GloVe embeddings yield negligible improvements across all architectures, the underlying reasons for this lack of impact could differ.

For baseline M1 and M2 architecture, the primary limitation is architecture rather than semantic. M1 compresses the visual input into a single global vector and the LSTM decoder lacks the visual context to effectively align with the rich, pre-existing semantic space provided by GloVe. M2 introduces spatial feature and attention mechanism, the performance from random to GloVe embeddings shows improvements in BLEU and remain marginal in the other metrics. Conversely, for the M3 architecture, the negligible difference between different embeddings indicates a limitation of data scale. Because the training set is relatively small, the transformer reaches the performance ceiling and begins to overfit as illustrated in the earlier training results. Therefore, it does not fully exploit the high-dimensional semantic spaces provided by the GloVe embeddings, and embeddings don't play a decisive factor in the results.

3.4. Qualitative Analysis

To further understand model behavior beyond quantitative metrics, we present qualitative comparisons across three levels of difficulty: easy, medium, and challenging cases, as shown in Figure 3.

In easy cases (Figure 3, left), all models generate nearly identical and accurate captions, correctly identifying the main object and action (e.g., dog and running in snow). This suggests that when visual cues are salient and unambiguous, architectural differences have minimal impact on performance. In medium cases (Figure 3, middle), all models correctly capture the main scene but differ in the level of detail. For example, while all models recognize a dog playing in water, M1 describes only the general activity ("playing in the water") and misses the key object (tennis ball). M2 captures the tennis ball but omits the water context. In contrast, M3 and M4 incorporate both the tennis ball and water, with M4 further including attribute-level details such as color. This demonstrates that models with stronger representation and decoding capacity produce more complete and descriptive captions. In challenging cases (Figure 3, right), all models fail to fully capture the ground-truth semantics, but exhibit distinct error patterns. For instance, none of the models detect the small object or the interaction (watching), instead generating coarse descriptions such as "people standing near water." M1 and M4 produce plausible but generic captions, M2 partially improves environmental understanding (e.g., "lake"), while M3 generates an unrelated scene (e.g., involving a building), indicating hallucination. These differences highlight limitations in capturing fine-grained details and interactions.

Overall, as task difficulty increases, model differences become more pronounced. Transformer-based models tend to produce richer descriptions in moderately complex scenarios but may also introduce instability in difficult cases. This behavior is likely amplified by the limited size of the dataset (Flickr8k), which can lead to reliance on frequent patterns and reduced robustness.

4. Discussion and Conclusion

Our controlled experiments with four different models reveals a fundamental tradeoff between complexity and representation power. The baseline M1 (Global+LSTM) is computationally efficient but lacks complexity and capacity, resulting in worst performance in the comparison. M2 adds complexity by using attention mechanisms to align text tokens with image local regions, and the performance sits in the middle ground. M3 (Spatial+Transformer) achieves the highest metrics but easy to overfit with small training data. M4 underperformed compared to M3 due to inefficient use of cross-attention with compressed global visual feature.

This project successfully evaluated four different architectures to isolate the impact of visual representation and decoding architecture in the image captioning task. The results clearly demonstrate that preserving spatial visual features is critical for accurate image-text alignment. Furthermore, with rich spatial features from encoder, the Transformer architecture outperforms traditional attention-LSTM approach, by utilizing masked self-attention and cross-attention to generate highly contextual captions for image.

[TODO] The CS 7643 rubric expects the report to also address deep-learning-specific reflection questions: how the model structure reflects the problem structure; which parts of the model have learned parameters and which do not; how the data is pre/post-processed; whether the model overfit and how it generalized; what hyperparameters mattered and how they were chosen; what framework was used; what existing code or models were used as starting points; and potential future work. v2 does not address these directly; add a short paragraph (or expand this section) to cover them.

Work Division

[TODO] Paragraph description not yet written. The rubric asks for both a short paragraph and the table below.

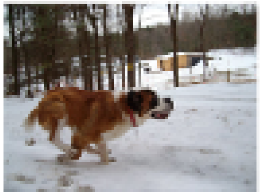
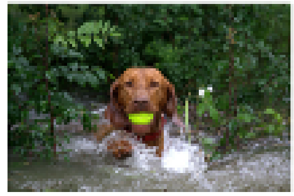
	Easy Case	Medium Case	Challenging Case
Image			
GT *(5)*	(1) A beautiful brown and white St Bernard running in the snow. (2) A St Bernard lunges through the snow. (3) Large St Bernard dog wearing red collar galloping through the snow. (4) St Bernard dog running in the snowy field. (5) The large brown and white dog is running through the snow .	(1) A brown dog is running through neck-deep water carrying a tennis ball. (2) A brown dog splashes in the water while carrying a ball in its mouth. (3) A brown dog swims through water outdoors with a tennis ball in its mouth. (4) A red dog, holding a ball, splashes through vegetation filled water. (5) Brown dog with tennis ball in mouth, in water and bushes .	(1) A child and a woman are at waters edge in a big city. (2) a large lake with a lone duck swimming in it with several people around the edge of it. (3) A little boy at a lake watching a duck. (4) A young boy waves his hand at the duck in the water surrounded by a green park. (5) Two people are at the edge of a lake, facing the water and the city skyline .
M1	**a brown and white dog is running through the snow.**	**a brown dog is playing in the water**	**two people are standing on a beach**
M2	**a brown and white dog is running through the snow.**	**a brown dog is playing with a tennis ball**	**three people are standing in the water near a lake**
M3	**a brown and white dog is running through the snow.**	**a brown dog is playing with a tennis ball in the water**	**a man in a black shirt is standing in front of a building**
M4	**a brown and white dog is running through the snow.**	**a brown and white dog is playing with a tennis ball in the water**	**two people are standing in front of a large body of water**

Figure 3. Qualitative comparison of generated captions across three difficulty levels (easy, medium, and challenging).

Members	Contributions
Hao Yang	Developed the implementation of the M3 model, beam search decoding, and evaluation metrics. Contributed to the design of the overall codebase structure and the unification of all four models (M1 to M4). Ran experiments for the M3 model. Initiated report organization and completed Abstract, Section 2, and Section 3.4.
Yongjian Wei	M2: Spatial encoder and LSTM attention code development. Run experiments for M2 model. Report writing section 1, section 3.2.
Qun Wei	Co-development of multi-layers LSTM attention decoder with Yongjian. Implemented Global encoder and a unified model for M1, M2, M3, and M4 for team collaboration. Run experiments for embeddings for all models. Report writing section 3.1, 3.3 and 4.
Oscar A. Martinez	Transcription of report from Word to LaTeX, and overall report formatting Initial M1 and M4 code development. Creation of Optuna scripts for hyperparameter tuning (M1 and M4; unused)

Table 7. Contributions of team members.

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